

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-aaef7623c32e82a28.jsonl`  
- SHA-256: `486b2b5f1538a72a89f2aa18cbe2bdeabaa48b2e5ced36630313c1fc8361522c`  
- Source modified: `2026-05-31T04:23:43+00:00`  
- Imported at: `2026-07-05T16:48:20+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:22:52.426Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables): TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"Surya's CODE (the repository LICENSE file at master) is now Apache License 2.0, a permissive license with no revenue restrictions or commercial-use caps. This means the source title's framing of 'GPL-3.0 code' is stale/outdated for the current master branch; only the weights carry the restrictive license."

****Source:**** <https://github.com/VikParuchuri/surya/blob/master/LICENSE> (primary)

****Supporting quote:**** "Apache License
Version 2.0, January 2004
<http://www.apache.org/licenses/>"

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:22:55.034Z)

I'll verify this claim about Surya's CODE license being Apache 2.0. Let me start by loading the web tools and checking the primary source.

3. assistant (2026-05-31T04:22:58.353Z)

I'll fetch the primary source first, and run searches for contradicting evidence in parallel.

4. user (2026-05-31T04:23:02.792Z)

License Analysis

****License Type:**** Apache License 2.0

****First 5 Lines (Verbatim):****

"Apache License
Version 2.0, January 2004
<http://www.apache.org/licenses/>
TERMS AND CONDITIONS FOR USE, REPRODUCTION, AND DISTRIBUTION
1. Definitions."

****Modifications or Exceptions:**** No. This is the standard Apache License 2.0 text with no additional clauses, modifications, or exceptions added. The file contains the complete, unaltered license as published by the Apache Software Foundation.

5. user (2026-05-31T04:23:07.138Z)

Web search results for query: "Surya OCR VikParuchuri license GPL-3.0 commercial weights 2025"

Links: [{"title": "GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub", "url": "https://github.com/CibiQuadance/surya-OCR"}, {"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "Surya OCR | tasarim.ai", "url": "https://tasarim.ai/en/models/surya-ocr"}, {"title": "surya/LICENSE at master · datalab-to/surya", "url": "https://github.com/VikParuchuri/surya/blob/master/LICENSE"}, {"title": "pitapo/surya · Hugging Face", "url": "https://huggingface.co/pitapo/surya"}, {"title": "GitHub - adithya-s-k/omniparse: Ingest, parse, and optimize any data format ?? from documents to multimedia ?? for enhanced compatibility with GenAI frameworks", "url": "https://github.com/adithya-s-k/omniparse"}, {"title": "GitHub - cognitiveailab/surya-sci: OCR, layout analysis, reading order, line detection in 90+ languages · GitHub", "url": "https://github.com/cognitiveailab/surya-sci"}, {"title": "View as Markdown", "url": "https://replicate.com/cudanexus/ocr-surya/llms.txt"}, {"title": "surya-ocr 0.1.2", "url": "https://pypi.org/project/surya-ocr/0.1.2/"}, {"title": "surya/pyproject.toml at master · datalab-to/surya", "url": "https://github.com/VikParuchuri/surya/blob/master/pyproject.toml"}]

Based on the search results, here's what I found about Surya OCR, Vik Paruchuri, the GPL-3.0 license, and commercial weights:

License and Commercial Usage

Surya OCR is released under the GPL-3.0 license and is open source and actively maintained with regular updates.

The project is licensed under "GPL-3.0-or-later" according to its configuration files.

Commercial Weights and Restrictions

Regarding commercial usage of the weights, the weights for the models are licensed cc-by-nc-sa-4.0, but there is a waiver for any organization under \$5M USD in gross revenue in the most recent 12-month period AND under \$5M in lifetime VC/angel funding raised.

Dual Licensing Option

If you want to remove the GPL license requirements for inference or use the weights commercially over the revenue limit, you can contact surya@vikas.sh for dual licensing.

2025 Development

According to recent citations, Surya is described as "A lightweight document OCR and analysis toolkit" with 2025 as the year, authored by Vikas Paruchuri and the Datalab Team.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. user (2026-05-31T04:23:07.671Z)

Web search results for query: "Surya OCR license change Apache GPL commercial use restrictions"

Links: [{"title": "GitHub - datalab-to/surya: OCR, layout analysis, reading order, table recognition in 90+ languages · GitHub", "url": "https://github.com/datalab-to/surya"}, {"title": "surya-ocr · PyPI", "url": "https://pypi.org/project/surya-ocr/"}, {"title": "GitHub - CibiQuadance/surya-OCR: OCR and line detection in 90+ languages · GitHub", "url": "https://github.com/CibiQuadance/surya-OCR"}, {"title": "surya/LICENSE at master · datalab-to/surya", "url": "https://github.com/VikParuchuri/surya/blob/master/LICENSE"}, {"title": "pitapo/surya · Hugging Face", "url": "https://huggingface.co/pitapo/surya"}, {"title": "Surya OCR | tasarim.ai", "url": "https://tasarim.ai/en/models/surya-ocr"}, {"title": "View as Markdown", "url": "https://replicate.com/cudanexus/ocr-surya/llms.txt"}, {"title": "Surya ? Try & Compare | Roboflow Playground", "url": "https://roboflow.com/model/surya"}, {"title": "Surya: The Revolutionary OCR Toolkit for 90+ Languages", "url": "https://converter.brightcoding.dev/blog/surya-the-revolutionary-ocr-toolkit-for-90-languages"}, {"title": "Apache Licensing and Distribution FAQ | Apache Software Foundation", "url": "https://www.apache.org/foundation/license-faq.html"}]

Based on the search results, here's what I found about Surya OCR's licensing:

Current Licensing Structure

The Surya code is licensed under Apache 2.0, while the model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue).

However, there appears to be some variation in different versions:

- The weights are licensed cc-by-nc-sa-4.0, but the creator will waive that for any organization under \$5M USD in gross revenue in the most recent 12-month period.

Commercial Use Restrictions

If you want to remove the GPL license requirements for inference or use the weights commercially over the revenue limit, you can contact the creator at surya@vikas.sh for dual licensing.

You also must not be competitive with the Datalab API.

Note on Apparent License Variations

The search results show that different repositories and documentation mention both Apache 2.0 and GPL-3.0 licensing, suggesting there may have been changes or different versions of the project. Surya was first released in January 2024 under the GPL-3.0 license, though current official sources cite Apache 2.0 for the code.

If you're looking for specific details about a recent license change, you may want to check the official GitHub repository directly for the most up-to-date information.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-05-31T04:23:11.512Z)

The primary source confirms the LICENSE file at master is standard Apache 2.0. But I need to verify the nuance in the claim. Note the repo moved from `VikParuchuri/surya` to `datalab-to/surya` (the blob URL redirects). Let me verify the current README and pyproject.toml to confirm the code-vs-weights distinction and check for any GPL remnants, since one search result said pyproject.toml lists "GPL-3.0-or-later".

8. user (2026-05-31T04:23:14.731Z)

The license field is set to Apache. Here is the exact line verbatim:

```
`license = { text = "Apache-2.0" }`
```

This indicates the project uses the Apache License 2.0.

9. user (2026-05-31T04:23:17.183Z)

Surya Licensing Summary

Based on the README's "Commercial usage" section:

(1) Code License: Apache 2.0

> "The Surya code is licensed under Apache 2.0."

(2) Model Weights License: Modified AI2 Open Rail-M

> "The model weights use a modified AI Pubs Open Rail-M license (free for research, personal use, and startups under \$5M funding/revenue)."

(3) Commercial Restrictions: Yes, there are revenue/funding caps and dual-licensing available.

The README states that model weights are free for research, personal projects, and startups with less than \$5 million in funding or revenue. For broader commercial licensing beyond these restrictions, Datalab offers paid licensing through their pricing page.

10. assistant (2026-05-31T04:23:28.168Z)

I now have strong primary-source confirmation. Let me assess the claim against the evidence.

What the claim asserts (three parts):

1. Surya's CODE (LICENSE file at master) is now Apache 2.0 ? **CONFIRMED** by the primary source (LICENSE file is verbatim standard Apache 2.0) and corroborated by `pyproject.toml` (`license = { text = "Apache-2.0" }`) and the README ("The Surya code is licensed under Apache 2.0").
2. The "GPL-3.0 code" framing is stale/outdated for current master ? **CONFIRMED**. Surya was originally GPL-3.0 (released Jan 2024), and search results explicitly note "Surya was first released in January 2024 under the GPL-3.0 license, though current official sources cite Apache 2.0 for the code." The current master is Apache 2.0.
3. "Only the weights carry the restrictive license" ? **CONFIRMED**. The weights use a modified AI Pubs Open Rail-M license with a \$5M revenue/funding cap and dual-licensing requirement; the code is clean Apache 2.0.

Adversarial checks:

- **Is the quote sufficient?** Yes ? the quote is the literal Apache 2.0 header from the primary LICENSE file, and the WebFetch confirms no modifications/exceptions were added.
- **Contradicting evidence?** The one apparent contradiction (a search result citing "GPL-3.0-or-later" in pyproject.toml) is **stale data** ? my direct fetch of the current `pyproject.toml` shows `Apache-2.0`. The GPL references reflect Surya's historical license and third-party forks (e.g., `CibinQuadance/surya-OCR`), not current master. This actually **supports** the claim's "stale/outdated" framing rather than refuting it.
- **Source quality?** Primary GitHub source, directly fetched. Matches the claim's strength.
- **Outdated?** No ? current master as of the fetch. The claim correctly identifies the OLD framing as the outdated one.
- **Marketing fluff?** No.

The claim is accurate and well-supported. One minor caveat: the claim is careful and correct in scope ("only the weights carry the restrictive license") ? it does not overreach by claiming the weights are usable. The weights ARE restrictive (non-commercial-ish, revenue-capped), which is consistent with everything found. The claim is verified.

11. user (2026-05-31T04:23:38.128Z)

Structured output provided successfully

12. assistant (2026-05-31T04:23:42.936Z)

I've completed the verification. The claim is ****not refuted**** ? it is well-supported by primary sources.

Summary of findings:

- ****Surya's LICENSE file at master IS standard Apache 2.0**** (verbatim, unmodified), confirmed by direct fetch and corroborated by the current ``pyproject.toml`` (``license = { text = "Apache-2.0" }``) and README.
- The ****"GPL-3.0 code" framing is genuinely stale**** ? that was Surya's original January 2024 license; current master is Apache 2.0.
- ****"Only the weights carry the restrictive license" is accurate**** ? the weights use a modified AI Pubs Open Rail-M license with a \$5M revenue/funding cap and dual-licensing requirement, while the code is clean Apache 2.0.
- The one apparent contradiction (a search snippet citing "GPL-3.0-or-later") is outdated/third-party fork data and actually reinforces the claim's "stale framing" point.

Confidence: high. Structured output submitted.